

Zero trust architecture for OT security in manufacturing, addressing brownfield legacy systems and NIS2.

Manufacturing × OT and industrial control system security × Zero trust architecture · OS096 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
74 is the world moving	86 can we play, can we win	MEDIUM 4 named in the evidence	€6.5bn observed evidence · per year	NEXT regulation adopted or propose...	L0 18 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

We help manufacturers secure their OT environments with a zero trust architecture that works with legacy brownfield systems, addressing NIS2 compliance and the risks of IT/OT convergence. This is urgent now because regulators and industry standards increasingly mandate zero trust, and brownfield sites cannot be simply replaced.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€9.0bn €1.6bn – €48.4bn	€6.5bn €1.2bn – €34.8bn	€388m €69.8m – €2.1bn	observed
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€1.3m €1.3m – €3.4m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Zero trust architecture	68.1 % of enterprises (10+ employees)	observed	Eurostat · isoc_cisce_ran2 · E_SECMNAC	2024
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 2 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

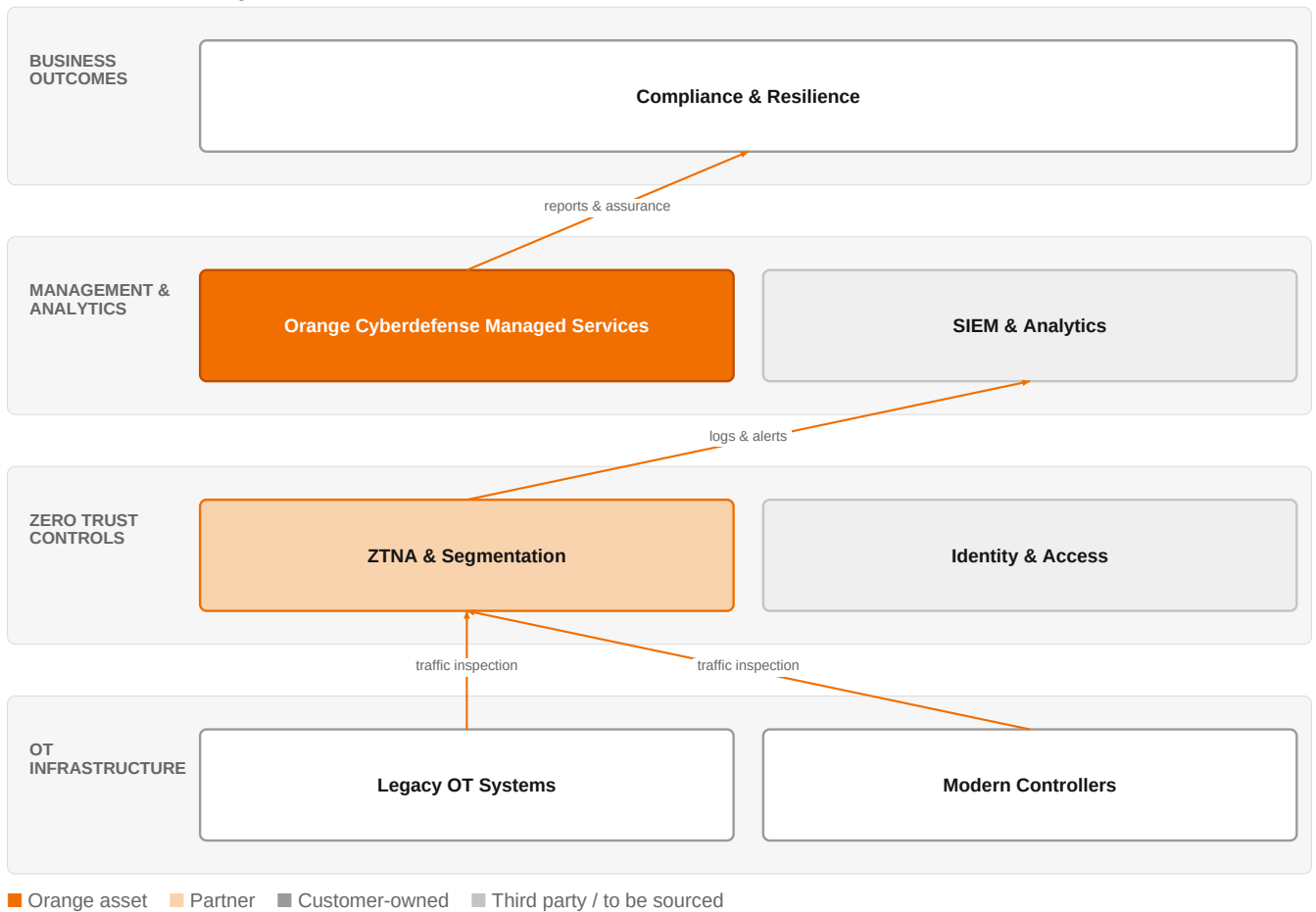
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 2 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

The solution, in one picture

Zero Trust OT Security Architecture



This architecture secures brownfield OT environments with zero trust controls, managed by Orange, to achieve compliance and operational resilience.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Regulatory pressure is mounting: NIS2 and other regulators now agree on zero trust as the security standard, making it non-negotiable for OT environments. IT/OT convergence is expanding the attack surface, and brownfield sites with legacy equipment cannot be replaced, so security must adapt to existing infrastructure. The market is moving from perimeter-based security to zero trust maturity, with a focus on visibility, segmentation, and operational resilience.

Sources: SIG-B3D9AD374F05 Atos 2026-05-06; SIG-022E77069770 ETCISO.in 2026-07-11; SIG-EE68A6085ADC Kings Research 2026-08-12; SIG-E2034B250974 bing.com 2026-08-13; SIG-164ECB40F03A TechTarget 2026-07-02; SIG-CFBD245A217D Industrial Cyber 2026-05-24

Who buys, and why now

The CISO or head of OT security signs, but the plant operations manager feels the pain of downtime and safety risks. The trigger is often a regulatory audit deadline (like NIS2) or a recent security incident that exposed legacy system vulnerabilities. They need to secure brownfield sites without disrupting production, and they need to demonstrate compliance to the board.

Why it is hot now — every claim bound to a dated source

- Brownfield industrial sites have a mix of legacy equipment that cannot be easily replaced, requiring security solutions that work with existing infrastructure. [SIG-E2034B250974]
- There is a growing trend of IT/OT convergence, bringing OT security into IT service management platforms, indicating a need for integrated security approaches. [SIG-117BA08606F7]
- Brownfield sites have a unique mix of legacy equipment and control systems that cannot simply be replaced, making zero-trust approaches necessary. [SIG-E2034B250974]

- IT/OT convergence brings OT security into platforms like ServiceNow, indicating a trend toward integrated OT security management. [SIG-117BA08606F7]
- Zero trust is becoming non-negotiable and there is specific guidance for OT environments. [SIG-022E77069770, SIG-574B25C143A3]
- Zero trust eliminates implicit trust at scale, applicable to industrial networks. [SIG-74A5048912D7]
- IT/OT convergence introduces both benefits and risks, requiring enhanced cybersecurity measures in industrial settings. [SIG-EE68A6085ADC]

What Orange would deliver, and why it can win

What Orange would deliver

We would start with an assessment using our cybersecurity experts to map the OT environment and identify critical assets and legacy systems. Then we would design a zero trust architecture leveraging our managed services and partnerships with Zscaler, Palo Alto Networks, and Fortinet for segmentation and access control. We would implement the solution in phases, using our ISO 27001 and TISAX certified processes to ensure compliance, and provide ongoing monitoring through Orange Cyberdefense managed services.

Why Orange can win

Orange can win because we combine deep cybersecurity expertise with a partner ecosystem that includes leading zero trust vendors. Our managed services and certifications (ISO 27001, TISAX) provide the trust and compliance assurance that manufacturers need. We also have experience with brownfield environments, as evidenced by our references in manufacturing.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Cyberdefense managed services	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
Palo Alto Networks	partner	L2	partner provides the required technology	unconfirmed
Fortinet	partner	L2	partner provides the required technology	unconfirmed
Zscaler	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 6.3; 4 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Zscaler	Security specialist	sells Zero trust architecture; competes in Cybersecurity — also an Orange partner	evidenced	SIG-46B6362BF559

Microsoft	Hyperscaler	named in this topic's evidence — also an Orange partner	evidenced	SIG-333EF44491D D
Atos / Eviden	Global systems integrator	competes in Cybersecurity	evidenced	SIG-B3D9AD374F 05
IBM	Global systems integrator	competes in Cybersecurity	evidenced	SIG-686BD502534 2
CrowdStrike	Security specialist	sells Zero trust architecture; competes in Cybersecurity	structural	—
Fortinet	Security specialist	sells Zero trust architecture; competes in Cybersecurity — also an Orange partner	structural	—
Palo Alto Networks	Security specialist	sells Zero trust architecture; competes in Cybersecurity — also an Orange partner	structural	—
Cisco	Network equipment vendor	sells Zero trust architecture; competes in Cybersecurity — also an Orange partner	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current OT security posture, and have you identified which legacy systems are most critical to protect?
- 2 Are you under regulatory pressure from NIS2 or similar frameworks, and what is your timeline for compliance?
- 3 Have you experienced any security incidents in your OT environment in the past 12 months?
- 4 How do you currently manage IT/OT convergence, and is there a single team responsible for both?
- 5 What is your tolerance for downtime during security implementation, and can we schedule changes during planned maintenance windows?

Objections you will meet

They will say	You can answer
We can't afford to replace our legacy systems, and zero trust seems like a big transformation.	That's a valid concern. Zero trust doesn't require replacement; it works with your existing infrastructure. We start with an assessment and implement in phases, minimizing disruption and focusing on the highest-risk areas first.
We already have a security vendor; why should we consider Orange?	Orange brings a holistic approach, combining our own managed services with best-of-breed partners. We don't lock you into one vendor; we design the right architecture for your specific brownfield environment, and we have the certifications and experience to ensure compliance.
Our OT systems are too sensitive to risk any disruption.	We understand. That's why we start with a thorough assessment and a phased implementation plan. We can schedule changes during maintenance windows and use non-intrusive monitoring to ensure minimal impact on production.

Risks and unknowns

The main risk is that brownfield OT environments are highly heterogeneous and fragile; any misstep could cause production downtime. There is also uncertainty about how quickly NIS2 enforcement will actually drive spending, and whether customers will accept a phased approach versus a big-bang transformation. We also need to validate that our partner solutions integrate seamlessly with legacy protocols and equipment.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a manufacturing client, using the Colgate-Palmolive, Mondelez International, Saint-Gobain Glass, Skytale, and Everise references to illustrate how Orange Cyberdefense managed services, backed by ISO 27001 and TISAX certifications, can deliver zero trust for OT while addressing brownfield constraints.
Sales	In a conversation with a manufacturing CISO, say: 'Given NIS2 and the challenge of securing legacy OT, we can bring together our managed security services with partners like Palo Alto Networks, Fortinet, or Zscaler to help you implement zero trust without ripping out your existing infrastructure.'
Strategist / Innovator	Commission a deep-dive brief on zero trust architecture for brownfield OT environments, leveraging Orange Group researchers and cybersecurity experts to map NIS2 compliance gaps and identify where Palo Alto Networks, Fortinet, or Zscaler capabilities can be integrated with Orange Cyberdefense managed services.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-2CC53D62856D	2025-08-18	openalex.org	1	A Hybrid Zero Trust Deployment Model for Securing O-RAN Architecture in 6G Networks
SIG-E2034B250974	2026-08-13	bing.com	2	SolutionsPT: Inside the new industrial cyber security playbook
SIG-EE68A6085ADC	2026-08-12	Kings Research	2	What is IT/OT Convergence? Benefits, Risks & Cybersecurity - Kings Research
SIG-117BA08606F7	2026-08-06	bing.com	2	IT/OT convergence brings OT security into ServiceNow
SIG-022E77069770	2026-07-11	ETCISO.in	2	Why Zero Trust is becoming non-negotiable - ETCISO.in
SIG-74A5048912D7	2026-07-10	Robotics & Automation News	2	How Zero Trust Network Access Eliminates Implicit Trust at Scale - Robotics & Automation News
SIG-164ECB40F03A	2026-07-02	TechTarget	2	Perimeter to posture: A roadmap to zero trust maturity - TechTarget
SIG-574B25C143A3	2026-06-26	csoonline.com	2	What CISOs need to tell the board about zero trust in OT: A 90-day communication and action pla...
SIG-5A3260810D62	2026-06-08	FedTech Magazine	2	Machine Identity Management: The Nonhuman Side of Federal Zero Trust - FedTech Magazine
SIG-686BD5025342	2026-05-28	IBM	2	Microsegmentation and zero trust: A natural fit - IBM
SIG-E94A28E305BA	2026-05-27	Network World	2	Zero trust isn't broken, but most companies are doing it wrong - Network World
SIG-CFBD245A217D	2026-05-24	Industrial Cyber	2	Zero trust in OT moves beyond identity as industrial operators prioritize visibility, segmentat...
SIG-D2DC2DA0537D	2026-05-14	appinventiv.com	2	Zero Trust Architecture Implementation in Australia - appinventiv.com
SIG-7E74F5F920F4	2026-05-12	EY	2	OT Control Tower - Zero Trust for Industrial Environments - EY
SIG-B3D9AD374F05	2026-05-06	Atos	2	NIS2, DORA, CISA, SAMA: Why Zero Trust Became the Security Standard Regulators Agree On - Atos
SIG-08B271ABAD4A	2026-04-30	iTnews	2	State of Security 2026: Zero Trust - iTnews
SIG-46B6362BF559	2026-04-30	IOT Insider	2	IXT integrates Zscaler Zero Trust into cellular IoT to replace VPNs and strengthen OT security ...
SIG-333EF44491DD	2026-04-29	Microsoft	2	Scale AI safely with Zero Trust security - Microsoft

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (named entities that were never supplied: Atos/Eviden)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T21:24:42+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.